

MLFF Physical Observable Validation Bridge Specification

mdstats project

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Scope

`mdstats.training_data.observable_validation` owns MLFF-specific pairing, statistical role, trajectory lineage, and later comparison decisions around analysis-owned observables. It does not implement or reinterpret the observable.

Advisory recommendation profile

`ObservableRecommendationProfile` lists currently executable baseline calls for generic condensed, crystalline, amorphous, liquid, and interface use cases. It is not the future compositional material-profile system. The `MaterialValidationProfile` name remains a compatibility alias only.

Verified collection identity

`ObservableCollectionIdentity` records frame semantics, dimensions, composition digest and counts, frame selection, geometry, dynamics, labels, provenance status, and source-content digests. Human labels and filesystem paths are location metadata and do not change scientific identity.

A supplied identity is never trusted. The bridge recomputes the identity from the collection actually passed to the analysis and rejects any mismatch. Object-dtype arrays are rejected because pointer bytes are not stable content. Large systems store composition counts and sequence digest without requiring the entire atomic-number sequence in evidence.

Symmetric trajectory-generation identity

`TrajectoryGenerationIdentity` is used for both reference and candidate trajectories. It binds:

- generator kind and artifact digest;
- optional generator manifest;
- calculation or MD protocol digest;
- exact output collection digest;
- engine and version;
- runtime environment;
- initial configuration and source provenance when available;
- precision policy, random seed, and notes.

The older `MLFFTrajectoryGenerationIdentity` is a compatibility wrapper. Under complete-lineage mode it must explicitly declare `output_collection_digest`. The bridge verifies both reference and candidate generation records against the actual collections.

Statistical role and activation

Every plan contains an `ObservableEvidenceRole`:

- `training_diagnostic`;
- `checkpoint_monitor`;
- `outer_validation`;
- `calibration`;
- `locked_test`;
- `external_benchmark`.

`ObservableValidationActivationRecord` binds partition policy/assignment, predeclared comparison-policy identity, protocol freeze, and locked-test activation as required by the role. Locked-test execution fails unless all required upstream identities are present.

The permitted order is:

```
comparison policy + activation + frozen protocol/partition identity
    -> observable execution evidence
    -> comparison result
    -> acceptance decision
```

Realized evidence must never tune its own thresholds or retroactively alter data selection, training protocol, checkpoint selection, calibration policy, or active-learning acquisition.

Plan and evidence

`MLFFObservableValidationPlan` binds the recipe, advisory profile, paired labels, lineage requirement, ionic-chain declaration, activation record, notes, and bridge version.

`MLFFObservableValidationEvidence` stores:

- verified reference/candidate collection identities;
- symmetric generation identities;
- runtime and capability identities;
- warnings and durations;
- paired native result types;
- paired analysis-owned `ObservableResultIdentity` records;
- activation and recipe/plan digests.

`MLFFObservableValidationEvidenceRecord` is a lightweight restorable, digest-verified summary. It does not load or duplicate scientific arrays.

Execution

```
run_mlff_observable_validation(  
    reference_collection,  
    candidate_collection,  
    plan,  
    reference_generation=...,  
    candidate_generation=...,  
)
```

Complete lineage is required by default. Diagnostic callers may explicitly disable the gate, but such evidence is not production validation evidence.

Ownership

The MLFF bridge owns recipe selection, pairing, lineage, role activation, comparison-policy identity, aggregation, and eventual checkpoint decisions. RDF, coordination, dynamics, transport, thermomechanical, energetic, and path algorithms remain under their own analysis architectures.

Comparison layer

Revision 0.20.52a0 implements the separate `ObservableComparisonPolicy`, `ObservableComparisonResult`, and `ObservableAcceptanceDecision` records in `mdstats.training_data.observable_comparison`. The bridge still owns only execution, pairing, and lineage; numerical discrepancy and checkpoint decisions remain downstream policy objects.

Required tests

Tests verify:

- supplied-identity mismatch rejection;
- symmetric complete lineage;
- output-collection binding;
- relocation-invariant identity;
- role and locked-test gates;
- analysis-owned result digests;
- evidence-record round trip and tamper rejection;
- recipe equality and paired runtime/capability lineage;
- no evidence-to-policy reverse dependency.